PROCTECH 4TR3 - Capstone Design Project II

Project Plan

6 ON 6IX

A Reimagined Volleyball Learning Application

Automation Systems Engineering Technology – Software Stream

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McMaster University | ENGINEERING

Project Statement

6 On Six is a mobile application designed to bridge the knowledge gap faced by new and intermediate volleyball players entering beginner and semi-competitive team environments. Inspired by Duolingo, we borrow the gamified microlearning model and apply it to volleyball, providing a structured platform for learning the fundamentals of the sport: rules, positions, terminology, rotations, and more advanced strategic concepts. As users progress, the complexity of the material scales, accordingly, allowing players to advance at their own pace and skill level.

The application is built to function entirely offline, with all learning content bundled at install time, so that it remains usable in gyms and practice facilities where connectivity is unreliable. By providing accessible, on-demand volleyball education, 6 On Six aims to accelerate individual player development and reduce the practice time teams lose to conceptual instruction.

Project Timeline

The timeline is presented as two schedules. Schedule 1 is our committed deliverable: a complete, fully offline application. Schedule 2 covers online features and App Store publication, which we will pursue only if Schedule 1 is complete and stable at the midterm checkpoint. Both schedules conclude by December 1, 2026.

Lead denotes the person accountable for a task being completed and correct. Support denotes a contributing member. Both members work in a single shared repository, so ownership indicates accountability rather than exclusive execution.

Schedule 1 – Committed Deliverable (Fully Offline Application)

Task	Deadline	Lead	Support
Foundation			
Finalize architecture and feature list	Sept 22	Issey	Yash
Scope lock: 4 units, 12 lessons, 80 questions, 50 glossary terms	Sept 22	Both	—
Shared repository and development conventions established	Sept 24	Issey	—
Content specification: question types, concept tagging, unit outline	Sept 26	Yash	Issey
Authoring pipeline: spreadsheet to JSON export with schema validation	Sept 29	Issey	Yash
Build			
Application shell: navigation, learning path, local storage	Oct 1	Issey	Yash
UI/UX direction: screen layouts and visual language	Oct 1	Yash	Issey
Unit 1 (Court, positions, and roles) content authored	Oct 6	Yash	Issey
Lesson player and question engine	Oct 6	Issey	Yash
Apple Developer Program enrolment submitted	Oct 6	Issey	—

Task	Deadline	Lead	Support
Court renderer, drag-and-drop rotation questions, haptic snap feedback	Oct 10	Issey	—
Progress tracking, streaks, and unit unlock logic	Oct 14	Issey	Yash
Unit 2 (Terminology and communication) content authored	Oct 15	Yash	Issey
Searchable glossary module	Oct 16	Issey	Yash
Offline verification on a physical device	Oct 18	Yash	Issey
Prototype freeze: question types frozen, demo-ready build	Oct 20	Issey	Yash
Midterm report and presentation materials	Oct 22	Yash	Issey
MIDTERM REPORT AND PRESENTATION	Oct 23	Both	—
Content Expansion and Validation			
Recruit participants, prepare sign-in sheet and test script	Oct 28	Yash	Issey
Unit 3 (Rotational fundamentals and serving order) content authored	Oct 31	Yash	Issey
Conduct user testing sessions	Nov 7	Yash	Issey
Unit 4 (The 5-1 system) content authored	Nov 7	Yash	Issey
Audio and haptic feedback pass, with independent settings toggles	Nov 7	Issey	Yash
Feedback synthesis and defect triage	Nov 10	Yash	Issey
Units 3 and 4 loaded and verified in application	Nov 10	Issey	Yash
End-of-unit review lessons built from concept tags	Nov 14	Issey	Yash
Implement revisions and bug fixes from testing	Nov 21	Issey	Yash
Conditional: units 5 and 6 authored and loaded	Nov 21	Yash	Issey
Release candidate: feature-frozen offline build	Nov 24	Issey	Yash
Final documentation, demo recording, presentation prep	Nov 28	Yash	Issey
PROJECT COMPLETE	Dec 1	Both	—

Schedule 2 – Extended Deliverables (Online Features and Publication)

Schedule 2 runs in parallel with the validation phase of Schedule 1, beginning only if the checkpoint below is passed. If any task slips more than five days past its listed deadline, Schedule 2 is stopped and the offline application is submitted as our final deliverable.

Task	Deadline	Lead	Support
CHECKPOINT: PROCEED WITH SCHEDULE 2?	**Oct 26**	**Both**	**—**
Backend project, database schema, and access policies	Oct 30	Issey	Yash
User accounts and authentication	Nov 3	Issey	Yash
In-app account deletion (App Store requirement)	Nov 5	Issey	Yash
Progress sync between device and server	Nov 7	Issey	—
Leaderboards	Nov 10	Issey	Yash
Asynchronous dual mode	Nov 13	Issey	Yash
TestFlight closed beta build distributed	Nov 14	Issey	Yash
Privacy policy, support URL, and App Privacy disclosures	Nov 14	Yash	Issey
App Store assets: icon, screenshots, description	Nov 15	Yash	Issey
Beta feedback collection and fixes	Nov 18	Both	—
SUBMIT TO APP STORE REVIEW	**Nov 20**	**Issey**	**Yash**
Respond to review feedback, resubmit if rejected	Nov 20 to 27	Issey	Yash
Public release	Nov 30	Issey	Yash
PROJECT COMPLETE	**Dec 1**	**Both**	**—**

Note: publication is scoped to iOS only. Publishing to the Google Play Store from a personal developer account requires a closed test with at least twelve testers opted in continuously for fourteen days before production access is granted, which is not achievable within this timeline.